How signers signal event boundaries

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Previous research has shown that telicity (whether an event has an endpoint) is used actively in language processing across spoken and sign languages. Even non-signing children and adults are able to distinguish between telic and atelic signs in ASL (American Sign Language), suggesting that humans may have an innate intuition about the boundedness of events [5, 10, 11, 14, 15]. While spoken languages don't use phonological features to signal telicity, previous studies have qualitatively identified four broad categories of ASL phonological features that suggest that the verb is telic: **A. change of orientation** (if the wrist is bent or straightened and if the palm is rotating downwards or upwards), **B. change of aperture** (closing/opening the hand), **C. setting change within a major body region** (if the sign moves across a body region), **D. change of location** (hand-to-body and between-hands contact) [2, 12]. Prior work has also shown that kinematic **motion information** (**M;** velocity, acceleration of hands) can also mark telicity [7]. To detect how these features are used in processing, our work defines measurable aspects of the phonological and motion ASL features and quantitatively examines whether/how they actually signify telicity. We operationalized the four phonological telic features into 17 metrics and the motion features into 8 metrics (25 metrics in total) [Table1]. We collected 100 signs from YouTube and SignBank videos. 50 of them were telic signs, and 50 were atelic, with some signs inspired by prior literature [3, 8, 9, 12,13]. Each sign is a single verb (phrase). All signs and their respective telicity labels were validated by a native ASL signer. We used the MediaPipe toolkit [6] to trace signers' movement across frames. From the movement traces, we extracted values of all 25 operational metrics for each video. We tested the discriminability of telicity according to our metrics using two methods to ensure replicability of our findings: Support Vector Machines (SVM) [4] and Locality Sensitive Hashing (LSH) [1]. While SVM requires training, LSH does not. The training of SVM finds the boundary that best separates data in terms of telicity, while the partitioned results of LSHs' random planes reflect which metrics are most influential and effective in separating the data [Figure1]. Due to space constraints, we only report the SVM results that are also supported by the LSH method. The SVM achieved 68% accuracy on five-fold cross-validation, and multivariate tests showed that each metrics group (A, B, C, D, M) was significantly helpful to the SVM ($p < 0.005$), confirming that phonological and motion features are both strong indicators of telicity. By likelihood ratio statistics, group M (motion) improves the model the most, followed by groups D (contact) and A (orientation). Within each feature, the most influential metrics are M9 (total movement path of the signer's whole body), A2 (how much the wrist bent/straightened), B3 (the standard deviation of how much hands open/close), C4 (directness of the path of the hand), and D2 (minimum distance between hand and body) [Figure 2]. Our work has quantitatively confirmed theoretical hypotheses from the literature, and proposed finer granularities of these features to better study them at scale. Our results show that the **Motion** features play a more prominent part than the four phonological features proposed in the literature, which suggests the need for a unifying framework to explain the interactions between phonological and motion features of ASL telicity expressions.

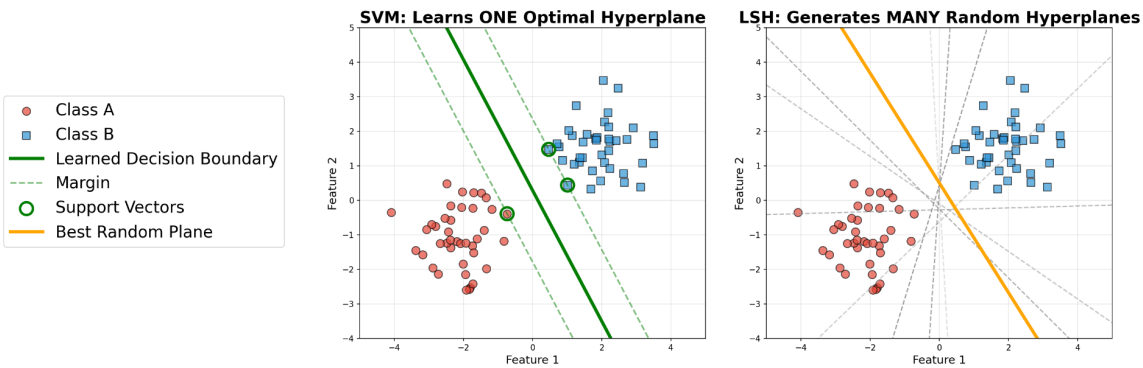

Figure 1: Illustration of SVM and LSH mechanism with a toy dataset. SVM uses gradient descent optimization, while LSH generates random partitions and selects the best one.

	A		B	C	D	
Phonological Features	Extension / Flexion (bending and straightening of the wrist)	Pronation / Supination (rotation of the wrist)	Aperture of the hand (open to close or close to open)	Uni-directional movement along x/y/z dimension of a body region	Contact between body and hand	Contact between two hands
	A1: max_angle **A2: sun_angle**	A3: max_rotation A4: sum_rotation	B1: aperture_change B2: aperture_range **B3: aperture_std**	C1: most_changed_region C2: most_changed_dim C3: most_changed_signed_value **C4: straightness**	D1:body_contact_frames **D2: min_body_distance** D3:body_contact_binary	D4: hand_contact_frames D5: min_hand_distance D6: hand_contact_binary
Motion Features	M1: mean_velocity, M2: velocity_std, M3: max_velocity, M4: min_velocity, M5: mean_acceleration, M6: acceleration_std, M7: max_acceleration, M8: min_acceleration, **M9: total_path_length**					

Table 1: Table of 25 operational metrics and their corresponding general features.

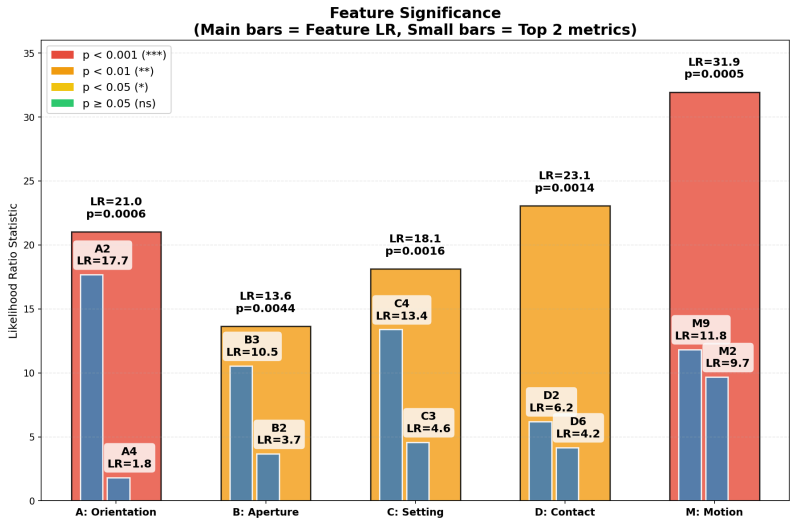

Figure 2: Feature significance and top 2 contributing metrics of each feature in SVM.

References: [1] Andoni & Indyk (2008). *Communications of the ACM*. [2] Brentari (1998). *MIT Press*. [3] Davidson et al. (2019). *Glossa*. [4] Hearst et al. (1998). *IEEE*. [5] Ji & Papafragou (2024). *Developmental Psychology*. [6] Lugaresi et al. (2019). *CVPR*. [7] Malaia & Wilbur (2012). *Language and speech*. [8] Rathmann (2003). *Proceedings of SULA*. [9] Rathmann (2006). The University of Texas at Austin. [10] Strickland et al. (2015). *Proceedings of the National Academy of Sciences*. [11] Wagner et al. (2024). *Academy of Sciences*. [12] Wilbur (2003). *Chicago Linguistic Society*. [13] Wilbur (2008). *Theoretical issues in sign language research.* [14] Krebs et al. (2023). *Language, Cognition and Neuroscience.* [15] Krebs et al. (2025) *Frontiers in Psychology.*